

An Immersive Labeling Method for Large Point Clouds

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Figure 1: Overview of Immersive labeling Method. A large point cloud of outdoor scene from Semantic3D rendered in VR(left-most). The user labels points using cube-shape selection primitive(middle). The 3D scene was labeled into multiple semantic parts(right-most). The user labels semantic information on points and delete corrupt points with instant visual feedback.

ABSTRACT

3D point clouds often require accurate labeling and semantic information. However, in the absence of fully automated methods, such labeling must be performed manually, which can prove extremely time and labour intensive. To address this, we propose a novel hybrid CPU/GPU-based algorithm allowing instantaneous selection and modification of points supporting very large point clouds. Our tool provides a palette of 3D interactions for efficient viewing, selection and labeling of points using head-mounted VR and controllers. We evaluate our method with 25 users on tasks involving large point clouds and find convincing results that support the use case of VR-based point cloud labeling.

Index Terms: Human-centered computing—Virtual Reality—Immersive interaction—Immersive labeling

1 INTRODUCTION

3D scans of real objects, structures and scenes are increasingly being employed in fields ranging from building virtual environments to training robots or autonomous cars to archaeology, museums, surveying and forensics, remote sensing and the creation of 3D assets for computer games and art. Current methods for capturing real world structures tend to produce dense, unstructured 3D point clouds which, before they can be used in their intended application, must often be classified into different subsets, be it by semantic meaning or spatial structure. However, the essential challenge facing the manual labeling of large point clouds is that this task is time-consuming, labour intensive and tedious. The majority of existing tools(e.g. Pointly.AI) are screen based, and so must present 3D data on a 2D display. With large, dense point clouds, 2D displays suffer from heavy occlusion, and making fast, accurate 3D selections of points can be both slow and error-prone. VR-based tools(e.g. ShootingLabels) demand for high resolution rendering at high frame rates. Often a trade-off between high resolution rendering and smooth interaction is faced.

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In this work, we propose a mixed CPU/GPU-based method that permits continuous level-of-detail rendering(CLOD rendering) and rapid selection, translation and removal of points with instantaneous visual feedback. Our method deploys a chunk-based compute shader selection algorithm fostering fast spatial queries and modification of point labels in large point clouds. In this way the efficiency of the labeling process is significantly improved through better of positioning and observation of details. By conducting a user study, we investigate if our immersive VR interface can accelerate this process. Depth perception may aid selection in 3D, and the ability to change viewpoint rapidly may help deal with heavy occlusion. To the best of our knowledge, our method is the first VR labeling application that supports large point clouds with only minor preprocessing while preserving the original color and geometry features of the point cloud.

2 METHOD

During the labeling process, our acceleration data structure successfully balances the rendering and interaction in Virtual Reality. The chunk based data structure costs very little on the CPU side, but it significantly reduces the workload of traversing point clouds. Users are fully immersed in the large-scale point cloud scene. The input and output of our method are point clouds in OBJ, PLY, or other binary formats. After reading, we use the Potree Converter in the adapted version of [2] to generate per point a single unsigned byte value representing the level of detail (LOD), and to split the points into chunks.

User Interface: We design a palette of 3D interactions for efficient viewing, selection and labeling of points using head-mounted VR and controllers as shown in Figure 1. We provide three "selection primitives" for painting labels onto points in 3D: *Sphere*, *Cube*, and *Clipping Plane*. The sphere and cube primitives are mainly designed for painting points closed to users. Simply pressing the trigger of the right controller will select all points currently within the active primitive. The size of the selection primitive is easily adjusted with the touchpad of the left controller. The clipping plane primitive is used for cutting points out of a wall or boundary and provides a method for selecting a large number of points rapidly.

Navigation in VR: Navigation is achieved in three ways. The user may teleport to any position in the point cloud using the left controller by shooting a ray to the desired location. Alternatively, the user can at any time simply "grab" the point cloud with the controllers and then scale, rotate or translate it with the relative motions of their hands. Finally, users can move in small increments

